LECTURE 5 · THE WEB AND HTTP

Computer Networks
Lecture 5 (Chapter 2, Lec 2)
The Web and HTTP

Status Codes | RTT | Web · URLs | TCP / UDP | HTTP

بنك الأسئلة

— QUESTIONS BANK —

ملخص البنك

Bank Summary

60	16	6
MCQ سؤال اختيار من متعدد	ESSAY سؤال مقالي	CALCULATION سؤال حسابي

توزيع الإجابة الصحيحة في الـ MCQ:

A = 16 | B = 12 | C = 14 | D = 18

مرتبة بحيث الـ high-likely يبجوا الأول في كل نوع

Part 1 — Multiple Choice Questions (MCQ)

.1

The Internet (and TCP/IP networks in general) makes how many transport protocols available to applications?

ترجمة: الإنترنت (وشبكات TCP/IP عموماً) بيوفّر كام بروتوكول نقل للتطبيقات؟

A. Four

B. One

C. Two

D. Three

الإجابة الصحيحة: C) Two

ليه صح: الإنترنت بيوفّر بروتوكولين للنقل بس: TCP و UDP. مزودش واحد ولا تلاتة، ده تعريف أساسي اتذكر أكتر من مرة.

المرجع: Lecture 2 — Section 2.1.4

Transport Layer TCP UDP ترجيح: high

.2

Which two transport protocols does the Internet provide to applications?

ترجمة: إيه البروتوكولين اللي الإنترنت بيوفّرهم للتطبيقات؟

A. HTTP and FTP

B. IP and ICMP

C. SMTP and POP3

D. TCP and UDP

الإجابة الصحيحة: D) TCP and UDP

ليه صح: الإجابة TCP و UDP. HTTP و FTP بروتوكولات تطبيق مش نقل، و IP بروتوكول شبكة مش نقل.

المرجع: Lecture 2 — Section 2.1.4

Transport Layer TCP UDP ترجيح: high

.3

The TCP service model includes which two main services?

ترجمة: نموذج خدمة TCP بيشمل أي خدمتين أساسيتين؟

A. Connectionless service and unreliable data transfer

B. Connection-oriented service and reliable data transfer service

C. Throughput guarantees and security

D. Congestion control and timing guarantees

الإجابة الصحيحة: B) Connection-oriented service and reliable data transfer service

ليه صح: TCP بيوفّر خدمة connection-oriented (بـ handshaking) وخدمة reliable data transfer. مفيش throughput/timing guarantees دي خدمات مش متوفرة في إنترنت النقل.

المرجع: Lecture 2 — Section 2.1.4 (TCP Services)

TCP Services ترجيح: high

.4

What is the handshaking procedure in TCP?

ترجمة: إيه إجراء الـ handshaking في TCP؟

A. Sending application messages without preparation

B. Compressing data before sending it

C. Exchanging transport-layer control information before application-level messages begin to flow

D. Encrypting the connection between client and server

الإجابة الصحيحة: C) Exchanging transport-layer control information before application-level messages begin to flow

ليه صح: الـ handshaking هو تبادل معلومات تحكم في طبقة النقل قبل ما تبدأ رسائل التطبيق. التشفير والضغط مش وظيفة الـ handshaking.

المرجع: Lecture 2 — Section 2.1.4 (Connection-oriented service)

TCP Handshaking Connection-oriented ترجيح: high

.5

After the handshaking phase, a TCP connection is said to exist between what?

ترجمة: بعد مرحلة الـ handshaking، بيتقال إن اتصال TCP موجود بين إيه؟

A. The applications themselves

B. The sockets of the two processes

C. The routers along the path

D. The IP addresses of the hosts

الإجابة الصحيحة: B) The sockets of the two processes

ليه صح: الاتصال بين sockets العمليتين (client و server). مش بين الراوترز ولا الـ IP addresses، ده تعريف مهم.

المرجع: Lecture 2 — Section 2.1.4 (Connection-oriented service)

TCP Sockets Connection ترجيح: high

.6

The TCP connection is described as which type of connection?

ترجمة: اتصال TCP بيتوصف كـ إيه؟

A. Simplex

B. Broadcast

C. Full-duplex

D. Half-duplex

الإجابة الصحيحة: C) Full-duplex

ليه صح: اتصال TCP full-duplex: العمليتين يقدروا يبعتوا لبعض في نفس الوقت. مش half-duplex (اتجاه واحد في كل مرة) ولا simplex.

المرجع: Lecture 2 — Section 2.1.4 (Connection-oriented service)

TCP Full-duplex ترجيح: high

.7

What must an application do when it finishes sending messages over a TCP connection?

ترجمة: إيه اللي لازم التطبيق يعمله لما يخلص بعت الرسائل على اتصال TCP؟

A. Leave the connection open indefinitely

B. Send a UDP packet to confirm

C. Restart the server process

D. Tear down the connection

الإجابة الصحيحة: D) Tear down the connection

ليه صح: لازم يـ tear down (يفك) الاتصال. سيبه مفتوح للأبد مش ممارسة صح، و UDP مالوش علاقة بإنهاء اتصال TCP.

المرجع: Lecture 2 — Section 2.1.4 (Connection-oriented service)

TCP | Connection teardown | ترجيح: high

.8

TCP's reliable data transfer service guarantees that the data sent:

ترجمة: خدمة النقل الموثوق في TCP بتضمن إيه للبيانات المبعوتة؟

A. Arrives faster than UDP

B. Arrives without error and in the proper order, with no missing or duplicate bytes

C. Arrives only in correct order, but may be duplicated

D. Arrives encrypted end-to-end

الإجابة الصحيحة: B) Arrives without error and in the proper order, with no missing or duplicate bytes

ليه صح: TCP بيضمن وصول البيانات من غير أخطاء وبالترتيب الصحيح وبدون فقد أو تكرار. التشفير مش شغلانته، والسرعة دي ميزة UDP.

المرجع: Lecture 2 — Section 2.1.4 (Reliable data transfer service)

TCP | Reliable data transfer | ترجيح: high

.9

What does the TCP congestion-control mechanism do?

ترجمة: ميكانيزم التحكم في الازدحام في TCP بيعمل إيه؟

A. Speeds up the sender at all times

B. Encrypts data at congested routers

C. Throttles a sending process when the network is congested between sender and receiver

D. Routes packets around congested areas

الإجابة الصحيحة: C) Throttles a sending process when the network is congested between sender and receiver

ليه صح: TCP بيقلل (throttle) سرعة المرسل لما الشبكة مزدحمة. مش بيسرّع المرسل، ولا بيفرض تشفير، ولا بيعيد التوجيه.

المرجع: Lecture 2 — Section 2.1.4 (Reliable data transfer service)

TCP Congestion Control ترجيح: high

.10

Whom does the TCP congestion-control mechanism primarily benefit?

ترجمة: ميكانيزم التحكم في الازدحام في TCP بيفيد مين أساساً؟

A. Only the communicating processes

B. Only the clients

C. Only the Web servers

D. The general welfare of the Internet, rather than the direct benefit of the communicating processes

الإجابة الصحيحة: D) The general welfare of the Internet, rather than the direct benefit of the communicating processes

ليه صح: TCP congestion-control بيخدم مصلحة الإنترنت العامة مش العمليات المتواصلة مباشرة. ده نقطة مهمة الدكتور بيشدد عليها — مش خدمة مباشرة للتطبيق.

المرجع: Lecture 2 — Section 2.1.4 (Reliable data transfer service)

TCP Congestion Control ترجيح: high

.11

UDP is best described as which kind of transport protocol?

ترجمة: UDP بيتوصف إزاي كبروتوكول نقل؟

A. A secure, encrypted transport protocol

B. A no-frills, lightweight transport protocol providing minimal services

C. A heavy, reliable, connection-oriented protocol

D. A congestion-aware transport protocol with handshaking

الإجابة الصحيحة: B) A no-frills, lightweight transport protocol providing minimal services

ليه صح: UDP بسيط وخفيف (no-frills, lightweight) بأدنى خدمات. مش موثوق ولا بيفرض handshake ولا بيعمل congestion control.

المرجع: Lecture 2 — Section 2.1.4 (UDP Services)

UDP Lightweight ترجيح: high

.12

UDP is connectionless, which means that:

ترجمة: UDP connectionless يعني إيه؟

A. UDP requires authentication before sending

B. Handshaking is required before communication

C. UDP cannot send any data over the network

D. No handshaking occurs before the two processes start to communicate

الإجابة الصحيحة: D) No handshaking occurs before the two processes start to communicate

ليه صح: مفيش handshaking قبل ما العمليتين يبدأوا التواصل. ده عكس TCP اللي بيطلب handshake. UDP بيبعت على طول.

المرجع: Lecture 2 — Section 2.1.4 (UDP Services)

UDP Connectionless ترجيح: high

.13

What does UDP provide regarding data transfer?

ترجمة: UDP بيوفّر إيه بخصوص نقل البيانات؟

A. Acknowledgment of every received packet

B. Guaranteed delivery of every message in correct order

C. An unreliable data transfer service

D. A reliable data transfer service

الإجابة الصحيحة: C) An unreliable data transfer service

ليه صح: UDP نقل غير موثوق (unreliable) — مفيش ضمان وصول الرسالة، ويمكن توصل خارج الترتيب. عكس TCP اللي موثوق.

المرجع: Lecture 2 — Section 2.1.4 (UDP Services)

UDP Unreliable ترجيح: high

.14

Does UDP include a congestion-control mechanism?

ترجمة: هل UDP فيه ميكانيزم للتحكم في الازدحام؟

A. Yes, UDP throttles senders during congestion just like TCP

B. No, but the network layer enforces congestion control for UDP

C. No, the sending side of UDP can pump data into the network layer at any rate it pleases

D. Yes, but only for packets larger than 1500 bytes

الإجابة الصحيحة: C) No, the sending side of UDP can pump data into the network layer at any rate it pleases

ليه صح: UDP مفيهوش congestion control — المرسل يقدر يضخ بيانات بأي معدل. ده فرق أساسي بينه وبين TCP اللي فيه throttling.

المرجع: Lecture 2 — Section 2.1.4 (UDP Services)

UDP Congestion Control ترجيح: high

.15

Which services are NOT provided by today's Internet transport protocols?

ترجمة: إيه الخدمات اللي بروتوكولات نقل الإنترنت مش بتوفّرها؟

A. Reliable data transfer and security

B. Encryption and decryption

C. Handshaking and connection setup

D. Throughput and timing guarantees

الإجابة الصحيحة: D) Throughput and timing guarantees

ليه صح: الإنترنت بيوفر reliable data transfer و security (عن طريق TLS)، لكن مفيش أي throughput ولا timing guarantees. نقطة أساسية اتذكرت في الـ summary.

المرجع: Lecture 2 — Section 2.1.4 (Services Not Provided)

Transport Services | Throughput | Timing | ترجيح: high

.16

Which set of applications typically uses TCP as their transport protocol?

ترجمة: إيه التطبيقات اللي بتستخدم TCP عادة؟

A. E-mail, Web, file transfer, and remote terminal access

B. Internet telephony applications only

C. Live video conferencing applications only

D. DNS queries only

الإجابة الصحيحة: A) E-mail, Web, file transfer, and remote terminal access

ليه صح: التطبيقات اللي محتاجة نقل موثوق زي البريد والويب ونقل الملفات والوصول للطرفية. التطبيقات الحساسة للزمن (telephony) بتفضل UDP.

المرجع: Lecture 2 — Section 2.1.4 (Figure 2.5)

TCP | Applications | ترجيح: high

.17

Why do developers of Internet telephony applications usually prefer to run them over UDP?

ترجمة: ليه مطورين تطبيقات الإنترنت تليفوني بيفضلوا UDP؟

A. To circumvent TCP's congestion-control mechanism and packet overheads

B. Because UDP is more secure than TCP

C. Because UDP provides reliable data transfer

D. Because UDP requires handshaking before sending

الإجابة الصحيحة: A) To circumvent TCP's congestion-control mechanism and packet overheads

ليه صح: عشان يتجاوزوا ميكانيزم التحكم في الازدحام في TCP وحجم الـ packet overheads. UDP مش موثوق ولا آمن أكتر من TCP.

المرجع: Lecture 2 — Section 2.1.4 (Figure 2.5 commentary)

UDP | Internet Telephony | ترجيح: high

.18

Why might Internet telephony applications be designed to use TCP as a backup if UDP communication fails?

ترجمة: ليه تطبيقات الإنترنت تليفوني ممكن تستخدم TCP كبديل لو UDP فشل؟

A. Because UDP is more unreliable than TCP for voice

B. Because many firewalls are configured to block most types of UDP traffic

C. Because TCP is faster than UDP

D. Because TCP has lower latency than UDP

الإجابة الصحيحة: B) Because many firewalls are configured to block most types of UDP traffic

ليه صح: عشان فايروولز كتير بتـ block معظم أنواع UDP traffic. مش لأن TCP أسرع أو أقل latency — العكس صحيح.

المرجع: Lecture 2 — Section 2.1.4 (Figure 2.5 commentary)

UDP | Firewalls | TCP fallback | ترجيح: high

.19

An application-layer protocol defines which of the following?

ترجمة: بروتوكول طبقة التطبيق بيحدد إيه؟

A. Only the transport protocol to use

B. The types of messages, syntax, semantics of fields, and rules for sending and responding to messages

C. Only the semantics of the fields

D. Only the syntax of the messages

الإجابة الصحيحة: B) The types of messages, syntax, semantics of fields, and rules for sending and responding to messages

ليه صح: بيحدد 4 حاجات: أنواع الرسائل، الـ syntax، الـ semantics، وقواعد متى وإزاي تبعت وترد. تعريف أساسي متكرر في المحاضرة.

المرجع: Lecture 2 — Section 2.1.5

Application-Layer Protocols | Definition | **ترجيح: high**

.20

The Web's application-layer protocol is:

ترجمة: بروتوكول طبقة التطبيق بتاع الـ Web هو إيه؟

A. DNS

B. FTP

C. SMTP

D. HTTP

الإجابة الصحيحة: D) HTTP

ليه صح: HTTP هو بروتوكول الـ Web. FTP للملفات، SMTP للإيميل، DNS لـ directory service.

المرجع: Lecture 2 — Section 2.1.5

HTTP | Web | **ترجيح: high**

.21

Which of the following application-layer protocols is proprietary and NOT in the public domain (RFCs)?

ترجمة: إيه البروتوكول اللي proprietary ومش متاح في RFCs؟

A. Skype's application-layer protocol

B. SMTP

C. HTTP

D. FTP

الإجابة الصحيحة: A) Skype's application-layer protocol

ليه صح: Skype بروتوكوله proprietary. HTTP و SMTP و FTP متاحين في RFCs للعموم.

المرجع: Lecture 2 — Section 2.1.5

Application-Layer Protocols | Proprietary | ترجيح: high

.22

A Web page (document) consists of which of the following?

ترجمة: صفحة الويب (document) بتتكون من إيه؟

A. A single HTML file only

B. A base HTML file and several referenced objects

C. Only the URL of the page

D. Only image files

الإجابة الصحيحة: B) A base HTML file and several referenced objects

ليه صح: صفحة الويب بتتكون من base HTML file وعدة objects موثقة. مش بس HTML ولا صور.

المرجع: Lecture 2 — Section 2.2.1

Web Page | Objects | ترجيح: high

.23

A URL has which two components?

ترجمة: الـ URL بيتكون من إيه (مكونين)؟

A. Browser type and server type

B. IP address and port number

C. Protocol and method

D. The hostname of the server and the object's path name

الإجابة الصحيحة: D) The hostname of the server and the object's path name

ليه صح: hostname بتاع السيرفر + path name بتاع الـ object. مثال: www.someSchool.edu (hostname) و /someDepartment/picture.gif (path).

المرجع: Lecture 2 — Section 2.2.1

URL Web ترجيح: high

.24

HTTP uses which underlying transport protocol?

ترجمة: HTTP بيستخدم إيه كبروتوكول نقل تحتية؟

A. IP

B. TCP

C. ICMP

D. UDP

الإجابة الصحيحة: B) TCP

ليه صح: HTTP بيشتغل فوق TCP. UDP مش موثوق، IP و ICMP بروتوكولات شبكة مش نقل.

المرجع: Lecture 2 — Section 2.2.1

HTTP TCP ترجيح: high

.25

Why does HTTP not need to worry about lost data or how TCP recovers from loss?

ترجمة: ليه HTTP مش محتاج يقلق على البيانات المفقودة؟

A. Because HTTP retransmits lost packets itself

B. Because TCP provides a reliable data transfer service to HTTP

C. Because HTTP encrypts all data

D. Because the network never loses HTTP packets

الإجابة الصحيحة: B) Because TCP provides a reliable data transfer service to HTTP

ليه صح: TCP بيوفّر reliable data transfer لـ HTTP، فمش شغله يتعامل مع الفقد. ده مثال على فايدة layered architecture.

المرجع: Lecture 2 — Section 2.2.1

HTTP TCP Layered architecture ترجيح: high

.26

HTTP is described as a stateless protocol because:

ترجمة: ليه HTTP بيتوصف بأنه stateless؟

A. HTTP only supports GET requests

B. HTTP cannot establish a connection to the server

C. HTTP does not use TCP as its transport

D. The HTTP server maintains no information about the clients

الإجابة الصحيحة: D) The HTTP server maintains no information about the clients

ليه صح: السيرفر ما بيبقاش عنده أي معلومات عن الـ clients. لو العميل طلب نفس الـ object مرتين، السيرفر بيبعته تاني لأنه نسي. تعريف مهم.

المرجع: Lecture 2 — Section 2.2.1

HTTP Stateless ترجيح: high

.27

What is the default port number for HTTP?

ترجمة: إيه البورت الافتراضي بتاع HTTP؟

A. 80

B. 21

C. 443

D. 25

الإجابة الصحيحة: A) 80

ليه صح: 80 هو البورت الافتراضي لـ HTTP. 21 لـ FTP، 25 لـ SMTP، 443 لـ HTTPS.

المرجع: Lecture 2 — Section 2.2.2 (Non-Persistent step 1)

HTTP Port ترجيح: high

.28

As of 2020, the majority of HTTP transactions take place over which version?

ترجمة: حتى 2020، أغلب معاملات HTTP بتحصل على أي نسخة؟

A. HTTP/1.1

B. HTTP/0.9

C. HTTP/1.0

D. HTTP/2

الإجابة الصحيحة: A) HTTP/1.1

ليه صح: HTTP/1.1 هو الأغلب (RFC 7230). HTTP/2 لسه بيزيد في الـ support.

المرجع: Lecture 2 — Section 2.2.1

HTTP versions ترجيح: high

.29

The two types of connections in HTTP are:

ترجمة: إيه نوعين الاتصالات في HTTP؟

A. Stateful and stateless connections

B. Single and multiple connections

C. Local and remote connections

D. Non-persistent connections and persistent connections

الإجابة الصحيحة: D) Non-persistent connections and persistent connections

ليه صح: non-persistent (كل request/response في اتصال TCP منفصل) و persistent (اتصال TCP واحد يفضل مفتوح). مقارنة أساسية.

المرجع: Lecture 2 — Section 2.2.2

HTTP Connections ترجيح: high

.30

RTT (Round-Trip Time) is defined as:

ترجمة: RTT (Round-Trip Time) بيتعرف إزاي؟

A. The time to send one packet only

B. The time it takes for a small packet to travel from client to server and then back to the client

C. The time to download a full Web page

D. The time for a server to process a request only

الإجابة الصحيحة: B) The time it takes for a small packet to travel from client to server and then back to the client

ليه صح: وقت باكت صغير يروح من العميل للسيرفر ويرجع. بيشمل propagation، queuing، processing delays. تعريف مهم للمسائل الحسابية.

المرجع: Lecture 2 — Section 2.2.2 (RTT definition)

RTT Delay ترجيح: high

.31

The first two parts of the three-way handshake take approximately:

ترجمة: الجزأين الأولانيين من الـ three-way handshake بياخدوا كام؟

A. Three RTTs

B. Half an RTT

C. Two RTTs

D. One RTT

الإجابة الصحيحة: D) One RTT

ليه صح: واحد (1 RTT) RTT. الجزء التالت (acknowledgment من العميل) بيبعت مع الـ HTTP request.

المرجع: Lecture 2 — Section 2.2.2 (Figure 2.7)

RTT | Three-way handshake | ترجيح: high

.32

For non-persistent HTTP, the total response time to fetch the base HTML file is roughly:

ترجمة: لـ non-persistent HTTP، إجمالي زمن الـ response لجلب الـ HTML file تقريباً إيه؟

A. Two RTTs plus the transmission time at the server

B. Three RTTs plus the transmission time

C. One RTT plus the transmission time

D. Half an RTT plus the transmission time

الإجابة الصحيحة: A) Two RTTs plus the transmission time at the server

ليه صح: 2 RTT (واحد لـ handshake + واحد لـ request/response) + زمن النقل على السيرفر. قانون مهم بييجي في المسائل.

المرجع: Lecture 2 — Section 2.2.2 (response time formula)

HTTP | RTT | Non-persistent | ترجيح: high

.33

If a Web page consists of a base HTML file and 10 JPEG images (11 objects total) and uses non-persistent HTTP, how many TCP connections are generated?

ترجمة: لو صفحة ويب فيها 1 HTML + 10 JPEG (11 object) وتستخدم non-persistent HTTP، هيتولد كام TCP connection؟

A. 10

B. 1

C. 11

D. 2

الإجابة الصحيحة: C) 11

ليه صح: 11 اتصال، واحد لكل object، لأن في non-persistent كل object بياخد اتصال منفصل. ده عيب أساسي للـ non-persistent.

المرجع: Lecture 2 — Section 2.2.2 (Non-persistent walk-through)

HTTP | Non-persistent | TCP connections | ترجيح: high

.34

The default mode of HTTP uses which type of connections?

ترجمة: الوضع الافتراضي لـ HTTP بيستخدم إيه؟

A. Non-persistent connections

B. UDP transport instead of TCP

C. Persistent connections with pipelining

D. Persistent connections without pipelining

الإجابة الصحيحة: C) Persistent connections with pipelining

ليه صح: persistent connections مع pipelining. عشان يقلل الـ RTTs والـ overhead على السيرفر.

المرجع: Lecture 2 — Section 2.2.2 (Persistent Connections)

HTTP | Persistent | Pipelining | ترجيح: high

.35

Pipelining in HTTP means that:

ترجمة: Pipelining في HTTP يعني إيه؟

A. Requests are sent one at a time, waiting for each response

B. Requests are sent only after the connection is closed

C. Requests for objects can be made back-to-back without waiting for replies to pending requests

D. Requests are sent in random order to balance load

الإجابة الصحيحة: C) Requests for objects can be made back-to-back without waiting for replies to pending requests

ليه صح: الطلبات بتيجي back-to-back من غير استنى الردود. ده بيقلل latency بشكل كبير في persistent connections.

المرجع: Lecture 2 — Section 2.2.2 (Persistent Connections)

HTTP | Pipelining | ترجيح: high

.36

The first line of an HTTP request message is called the:

ترجمة: أول سطر في رسالة HTTP request بيتسمى إيه؟

A. Status line

B. Entity body

C. Request line

D. Header line

الإجابة الصحيحة: C) Request line

ليه صح: الـ request line هو في الـ response، و الـ status line بيجي بعد الـ request line.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Request | Request line | ترجيح: high

.37

Which is the most commonly used HTTP request method?

ترجمة: إيه أكتر method شائعة في HTTP request؟

A. GET

B. PUT

C. POST

D. DELETE

الإجابة الصحيحة: A) GET

ليه صح: GET هي الأكثر استخداماً لما البراوزر بيطلب object في الـ URL. POST للـ forms، PUT لرفع، DELETE للحذف.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods GET ترجيح: high

.38

The GET method is used when:

ترجمة: متى بيتستخدم الـ GET method؟

A. The user uploads a file to a server

B. The user fills out a form with sensitive data

C. The user deletes an object on the server

D. The browser requests an object, with the requested object identified in the URL field

الإجابة الصحيحة: D) The browser requests an object, with the requested object identified in the URL field

ليه صح: لما البراوزر يطلب object بيحدده الـ URL. الـ forms الحساسة بتستخدم POST، رفع الملفات PUT، الحذف DELETE.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods GET ترجيح: high

.39

The POST method is typically used when:

ترجمة: متى بيتستخدم الـ POST method؟

A. The user fills out a form, e.g., provides search words to a search engine

B. The browser requests a JPEG image

C. The user deletes a server object

D. The server requests data from the client

الإجابة الصحيحة: A) The user fills out a form, e.g., provides search words to a search engine

ليه صح: لما اليوزر يملا form (زي كلمات البحث في محرك بحث). الـ entity body بيشيل المدخلات.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods | POST | ترجيح: high

.40

The Host: header line in an HTTP request specifies:

ترجمة: الـ Host: header في HTTP request بيشير لإيه؟

A. The host on which the object resides

B. The preferred language of the user

C. The connection timeout value

D. The browser type making the request

الإجابة الصحيحة: A) The host on which the object resides

ليه صح: الـ host اللي عليه الـ object. مهم لـ Web proxy caches رغم إن اتصال TCP موجود بالفعل.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Headers | Host | ترجيح: high

41.

The Connection: close header line in a request tells the server that:

ترجمة: الـ Connection: close header في الـ request بيقول للسيرفر إيه؟

A. The browser wants the server to close the connection after sending the requested object

B. The client wants to upgrade to HTTP/2

C. The client wants a persistent connection

D. The client is about to close the browser

الإجابة الصحيحة: A) The browser wants the server to close the connection after sending the requested object

ليه صح: العميل عايز السيرفر يقفل الاتصال بعد ما يبعت الـ object — يعني مش عايز persistent. عكس persistent الافتراضي.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Headers | Connection | ترجيح: high

42.

An HTTP response message has how many sections?

ترجمة: رسالة HTTP response فيها كام قسم؟

A. Four

B. Two

C. Three (an initial status line, header lines, and the entity body)

D. One

الإجابة الصحيحة: C) Three (an initial status line, header lines, and the entity body)

ليه صح: 3: status line, header lines, entity body. الـ entity body بيشيل الـ object نفسه.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Response | Structure | ترجيح: high

.43

The status line of an HTTP response message has how many fields?

ترجمة: الـ status line في HTTP response فيها كام field؟

A. Four

B. Two

C. Three (protocol version, status code, status message)

D. One

الإجابة الصحيحة: C) Three (protocol version, status code, status message)

ليه صح: 3: protocol version, status code, status message. مثال: HTTP/1.1 200 OK.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Response | Status line | ترجيح: high

.44

What does the HTTP status code 200 OK mean?

ترجمة: إيه معنى status code 200 OK؟

A. The requested object has been permanently moved

B. The requested document does not exist

C. The request succeeded and the information is returned in the response

D. The request could not be understood by the server

الإجابة الصحيحة: C) The request succeeded and the information is returned in the response

ليه صح: الطلب نجح والمعلومات (الـ object) بترجع في الـ response. أكتر كود شائع.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes | 200 OK | ترجيح: high

.45

What does the HTTP status code 301 Moved Permanently mean?

ترجمة: إيه معنى status code 301 Moved Permanently؟

A. The requested object has been permanently moved; the new URL is specified in the Location: header

B. The request could not be understood

C. The HTTP protocol version is not supported

D. The request succeeded, info returned

الإجابة الصحيحة: A) The requested object has been permanently moved; the new URL is specified in the Location: header

ليه صح: الـ object اتنقل بشكل دائم، والـ URL الجديد في الـ Location: header. العميل بيجيبه أوتوماتيك.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes | 301 | **ترجيح: high**

.46

What does the HTTP status code 400 Bad Request mean?

ترجمة: إيه معنى status code 400 Bad Request؟

A. The object was permanently moved

B. The request succeeded

C. The requested document does not exist

D. A generic error code indicating that the request could not be understood by the server

الإجابة الصحيحة: D) A generic error code indicating that the request could not be understood by the server

ليه صح: خطأ عام إن السيرفر ما فهمش الطلب. مش نجاح ولا نقل ولا Not Found.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes | 400 | **ترجيح: high**

.47

What does the HTTP status code 404 Not Found mean?

ترجمة: إيه معنى status code 404 Not Found؟

A. The requested document does not exist on this server

B. The request succeeded

C. The HTTP version is not supported

D. The server is down for maintenance

الإجابة الصحيحة: A) The requested document does not exist on this server

ليه صح: الـ document مش موجود على السيرفر ده. مش النجاح ولا السيرفر down ولا version unsupported.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes | 404 | ترجيح: high

.48

What does the HTTP status code 505 HTTP Version Not Supported mean?

ترجمة: إيه معنى status code 505 HTTP Version Not Supported؟

A. The requested document does not exist

B. The requested HTTP protocol version is not supported by the server

C. The requested object has been permanently moved

D. The request succeeded, info returned

الإجابة الصحيحة: B) The requested HTTP protocol version is not supported by the server

ليه صح: نسخة HTTP اللي العميل بيطلبها مش مدعومة من السيرفر. غالباً عشان عميل بيطلب HTTP/2 على سيرفر قديم.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes | 505 | ترجيح: high

.49

The Last-Modified: header line in an HTTP response is critical for:

ترجمة: الـ Last-Modified: header في الـ response مهم لإيه؟

A. Object caching, both in the local client and in network cache servers (proxy servers)

B. Encrypting the connection

C. Specifying the browser type

D. Indicating the request method

الإجابة الصحيحة: A) Object caching, both in the local client and in network cache servers (proxy servers)

ليه صح: caching محلي و في proxy servers. بيشير لمتى آخر تعديل للـ object — ده اللي بيخلي الـ cache يحدد لو الـ object اتغير.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Headers | Last-Modified | Caching | ترجيح: high

.50

Transport protocol services are organized along how many dimensions in this section?

ترجمة: خدمات بروتوكولات النقل متقسمة كام بعد في القسم ده؟

A. Four

B. Three

C. Two

D. Five

الإجابة الصحيحة: A) Four

ليه صح: 4 أبعاد: reliable data transfer, throughput, timing, security. نقطة مهمة الدكتور بيذكرها في الـ summary.

المرجع: Lecture 2 — Section 2.1.4 (Services Not Provided)

Transport Services | Dimensions | ترجيح: medium

.51

How many important network applications does the book focus on in this chapter?

ترجمة: الكتاب بيركز على كام تطبيق شبكة مهم في الفصل ده؟

A. Six

B. Four

C. Three

D. Five

الإجابة الصحيحة: D) Five

ليه صح: 5 تطبيقات: الـ Web، البريد الإلكتروني، P2P، DNS، والبث المرئي. نقطة مرقمة في القسم 2.1.6.

المرجع: Lecture 2 — Section 2.1.6

Network Applications ترجيح: medium

.52

The original version of HTTP is called:

ترجمة: النسخة الأصلية من HTTP بتتسمى إيه؟

A. HTTP/0.9

B. HTTP/1.1

C. HTTP/2

D. HTTP/1.0

الإجابة الصحيحة: D) HTTP/1.0

ليه صح: HTTP/1.0 هو الأصلي (RFC 1945، early 1990s). HTTP/1.1 الأغلب حالياً، HTTP/2 الجديد.

المرجع: Lecture 2 — Section 2.2.1

HTTP versions ترجيح: medium

.53

The three-way handshake used to establish a TCP connection involves how many TCP segments?

ترجمة: الـ three-way handshake بيشمل كام TCP segment؟

A. Two

B. Three

C. Four

D. One

الإجابة الصحيحة: B) Three

ليه صح: client 3 segments: يبعت، server يرد بـ acknowledge، client يـ acknowledge تاني. ده عشان كده اسمه three-way.

المرجع: Lecture 2 — Section 2.2.2 (Figure 2.7)

TCP | Three-way handshake | ترجيح: medium

.54

The request line of an HTTP request message has how many fields?

ترجمة: الـ request line فيها كام field؟

A. Two

B. Four

C. One

D. Three

الإجابة الصحيحة: D) Three

ليه صح: 3: method field (GET/POST/...)، URL field، HTTP version field. تعريف مهم.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Request | Request line | ترجيح: medium

.55

The HEAD method is often used by application developers for:

ترجمة: الـ HEAD method بيتستخدم لإيه غالبا؟

A. Uploading files

B. Deleting objects

C. Encrypting the connection

D. Debugging

الإجابة الصحيحة: D) Debugging

ليه صح: debugging. مش لرفع أو حذف أو تشفير، ده شغل المطورين.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods | HEAD | ترجيح: medium

.56

The PUT method allows a user to:

ترجمة: الـ PUT method بيتيح للمستخدم يعمل إيه؟

A. Request a French version of an object

B. Close the TCP connection

C. Delete an object on the server

D. Upload an object to a specific path (directory) on a specific Web server

الإجابة الصحيحة: D) Upload an object to a specific path (directory) on a specific Web server

ليه صح: رفع object لمسار محدد على السيرفر. DELETE للحذف، Accept-language للغة، Connection: close لقفل الاتصال.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods | PUT | ترجيح: medium

.57

The User-agent: header line specifies:

ترجمة: الـ User-agent: header بيشير لإيه؟

A. The size of the requested object

B. The host name of the server

C. The state of the connection

D. The user agent, i.e., the browser type making the request

الإجابة الصحيحة: D) The user agent, i.e., the browser type making the request

ليه صح: نوع البراوزر (زي Mozilla/5.0). مفيد عشان السيرفر يبعت نسخ مختلفة من نفس الـ object لأنواع مختلفة.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Headers | User-agent | ترجيح: medium

.58

The Accept-language: header line indicates:

ترجمة: الـ Accept-language: header بيشير لإيه؟

A. The user prefers to receive a particular language version of the object, if available

B. The request method being used

C. The server software version

D. The browser type making the request

الإجابة الصحيحة: A) The user prefers to receive a particular language version of the object, if available

ليه صح: تفضيل لغة اليوزر (زي fr للفرنسية). لو موجودة السيرفر بيبعتها، غير كده بيبعت النسخة الافتراضية. ده content negotiation header.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Headers | Accept-language | ترجيح: medium

.59

The Content-Length: header line indicates:

ترجمة: الـ Content-Length: header بيشير لإيه؟

A. The number of bytes in the object being sent

B. The time when the object was created

C. The server's name

D. The type of the object in the entity body

الإجابة الصحيحة: A) The number of bytes in the object being sent

ليه صح: عدد البايتس في الـ object. Last-Modified للوقت، Content-Type للنوع، Server لاسم السيرفر.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Headers | Content-Length | ترجيح: medium

.60

The Content-Type: header line indicates:

ترجمة: الـ Content-Type: header بيشير لإيه؟

A. The number of bytes in the object

B. The browser type making the request

C. The type of the object in the entity body (e.g., HTML text)

D. The server's hostname

الإجابة الصحيحة: C) The type of the object in the entity body (e.g., HTML text)

ليه صح: نوع الـ object (مثلاً text/html). Content-Length للحجم، User-agent للبراوزر، Host للسيرفر.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Headers | Content-Type | ترجيح: medium

Part 2 — Essay Questions

Essay 1.

Compare and contrast the services provided by TCP and UDP to applications.

ترجمة: قارن وبيّن الفروق بين الخدمات اللي TCP و UDP بيوفّروهاها للتطبيقات.

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are the two transport protocols the Internet provides to applications.
TCP service model includes:
1. Connection-oriented service: client and server exchange control information via handshaking before application messages flow; the connection is full-duplex and must be torn down after use.
2. Reliable data transfer: TCP delivers all data without error, in proper order, with no missing or duplicate bytes.
3. Congestion-control mechanism: throttles sender when network is congested, benefiting the Internet as a whole.

UDP is a no-frills, lightweight protocol:
1. Connectionless: no handshaking before communication.
2. Unreliable data transfer: no guarantee messages will reach the receiving process; messages may arrive out of order.
3. No congestion-control mechanism: the sender can pump data into the network layer at any rate.

Key differences: TCP is reliable and connection-oriented; UDP is fast and connectionless. TCP is used for e-mail, Web, and file transfer; UDP is preferred for Internet telephony and other applications tolerant of some loss.

المرجع: Lecture 2 — Section 2.1.4 (TCP Services, UDP Services)

TCP UDP Comparison ترجيح: high

Essay 2.

Explain TCP's connection-oriented service in detail.

ترجمة: اشرح خدمة الـ connection-oriented في TCP بالتفصيل.

TCP's connection-oriented service requires the client and server to exchange transport-layer control information with each other before application-level messages begin to flow.
This handshaking procedure alerts the client and server, allowing them to prepare for an incoming onslaught of packets.
After the handshaking phase, a TCP connection is said to exist between the sockets of the two processes.
The connection is full-duplex, meaning both processes can send messages to each other over the connection at the same time.
When the application finishes sending messages, it must explicitly tear down the connection.
Connection-oriented service is implemented in detail in Chapter 3.

المرجع: Lecture 2 — Section 2.1.4 (Connection-oriented service)

TCP | Connection-oriented | Handshaking | **ترجيح: high**

Essay 3.

Explain TCP's reliable data transfer service and what it guarantees to the application.

ترجمة: اشرح خدمة النقل الموثوق في TCP وإيه اللي بتضمنه للتطبيق.

TCP's reliable data transfer service guarantees that the communicating processes can rely on TCP to deliver all data sent without error and in the proper order.
When one side of the application passes a stream of bytes into a socket, it can count on TCP to deliver the same stream of bytes to the receiving socket, with no missing or duplicate bytes.
This means applications built on TCP do not need to worry about lost packets, reordering, or duplication — TCP handles all of that internally.
TCP also includes a congestion-control mechanism, but this is a service for the general welfare of the Internet rather than for the direct benefit of the communicating processes.

المرجع: Lecture 2 — Section 2.1.4 (Reliable data transfer service)

TCP | Reliable data transfer | **ترجيح: high**

Essay 4.

What services are NOT provided by today's Internet transport protocols, and how do time-sensitive applications cope with this limitation?

ترجمة: إيه الخدمات اللي بروتوكولات نقل الإنترنت مش بتوفّرها، وإزاي التطبيقات الحساسة للزمن بتتعامل مع القصر ده؟

Transport protocol services are organized along four dimensions: reliable data transfer, throughput, timing, and security.
TCP provides reliable end-to-end data transfer, and TCP can be enhanced at the application layer with TLS to provide security.
However, today's Internet transport protocols conspicuously do NOT provide any throughput or timing guarantees.
Despite this, time-sensitive applications such as Internet telephony can run on the Internet because they have been designed to cope, to the greatest extent possible, with this lack of guarantee.
These applications often work fairly well, but the Internet cannot provide any timing or throughput guarantees to them — the service is best-effort, not guaranteed.

المرجع: Lecture 2 — Section 2.1.4 (Services Not Provided)

Transport Services | Throughput | Timing | **ترجيح: high**

Essay 5.

Define an application-layer protocol and list what it defines.

ترجمة: عرّف بروتوكول طبقة التطبيق واكتب اللي بحدده.

An application-layer protocol defines how an application's processes, running on different end systems, pass messages to each other.
In particular, an application-layer protocol defines:
1. The types of messages exchanged, for example request messages and response messages.
2. The syntax of the various message types, such as the fields in the message and how the fields are delineated.
3. The semantics of the fields, i.e., the meaning of the information in the fields.
4. Rules for determining when and how a process sends messages and responds to messages.
Some application-layer protocols (such as HTTP) are specified in RFCs and are therefore in the public domain; others (such as Skype's) are proprietary and not publicly available.

المرجع: Lecture 2 — Section 2.1.5

Application-Layer Protocols | Definition | **ترجيح: high**

Essay 6.

Distinguish between a network application and an application-layer protocol, with examples.

ترجمة: بيّن الفرق بين network application و application-layer protocol، مع أمثلة.

A network application is a complete system that provides a service to users; an application-layer protocol is only one piece (albeit a very important piece) of a network application.
Example 1 - The Web: The Web is a client-server application that allows users to obtain documents on demand. It includes a standard for document formats (HTML), Web browsers (Chrome, Internet Explorer), Web servers (Apache, Microsoft servers), and the HTTP application-layer protocol. HTTP defines only the format and sequence of messages exchanged between browser and Web server.
Example 2 - Netflix: Netflix's video service includes servers that store and transmit videos, other servers that manage billing and other client functions, clients (the Netflix app on your smartphone, tablet, or computer), and the DASH application-layer protocol. DASH defines only the format and sequence of messages exchanged between a Netflix server and client. Thus, an application-layer protocol is one component of the larger network application.

المرجع: Lecture 2 — Section 2.1.5 (Examples paragraph)

Network Applications | Application-Layer Protocols | Comparison | **ترجيح:** high

Essay 7.

Explain the key characteristics of HTTP, including its architecture, transport protocol, default port, and state.

ترجمة: اشرح الخصائص الأساسية لـ HTTP، بما فيها الـ architecture، البروتوكول التحتي، البورت الافتراضي، والـ state.

HTTP (Hyper Text Transfer Protocol) is the Web's application-layer protocol, defined in RFCs 1945, 7230, and 7540.
Key characteristics:
1. Client-server architecture: client program (browser) and server program execute on different end systems and talk to each other by exchanging HTTP messages. The Web server is always on, with a fixed IP address, and services requests from potentially millions of different browsers.
2. Uses TCP as its underlying transport protocol (not UDP). The HTTP client first initiates a TCP connection with the server.
3. Default port number is 80.
4. Stateless: the server maintains no state information about the client. If a client asks for the same object twice in a few seconds, the server resends the object because it has forgotten the previous request.
5. Original version is HTTP/1.0 (early 1990s); as of 2020, the majority of HTTP transactions take place over HTTP/1.1; HTTP/2 is increasingly supported.

المرجع: Lecture 2 — Section 2.2.1

HTTP | Architecture | TCP | Port 80 | Stateless | ترجيح: high

Essay 8.

Define the terms Web page, object, base HTML file, and URL, and explain how they are related.

ترجمة: عرّف المصطلحات: Web page، object، base HTML file، URL، واشرح العلاقة بينهم.

A Web page (also called a document) consists of objects.
An object is a file — such as an HTML file, a JPEG image, a JavaScript file, a CSS style sheet file, or a video clip — that is addressable by a single URL.
Most Web pages consist of a base HTML file and several referenced objects.
For example, if a Web page contains HTML text and five JPEG images, then the Web page has six objects: the base HTML file plus the five images.
The base HTML file references the other objects in the page with the objects' URLs.
Each URL has two components: the hostname of the server that houses the object, and the object's path name.
Example: the URL www.someSchool.edu/someDepartment/picture.gif has www.someSchool.edu as the hostname and /someDepartment/picture.gif as the path name.

المرجع: Lecture 2 — Section 2.2.1

Web Page | Objects | Base HTML | URL | ترجيح: high

Essay 9.

Explain why HTTP is called a stateless protocol and what this means for the server.

ترجمة: اشرح ليه HTTP بيتوصف بأنه stateless وإيه معنى ده للسيرفر.

HTTP is called a stateless protocol because the server sends requested files to clients without storing any state information about the client.
If a particular client asks for the same object twice in a period of a few seconds, the server does not respond by saying that it just served the object to the client — instead, the server resends the object because it has completely forgotten what it did earlier.
Because an HTTP server maintains no information about clients, HTTP is said to be a stateless protocol.
This design simplifies server implementation and allows the server to handle many clients simultaneously without per-client state overhead.
However, Web applications that need to track user behavior (e.g., shopping carts, login sessions) must add state on top of HTTP using mechanisms such as cookies — which are not part of the core HTTP protocol discussed here.

المرجع: Lecture 2 — Section 2.2.1 (stateless paragraph)

HTTP | Stateless | ترجيح: high

Essay 10.

Compare non-persistent and persistent HTTP connections, mentioning the advantages of persistent connections.

ترجمة: قارن بين اتصالات non-persistent و persistent في HTTP، واذكر مزايا الـ persistent.

Non-persistent connections: each request/response pair is sent over a separate TCP connection. Each TCP connection transports exactly one request message and one response message, and the connection is closed after the server sends the object. Persistent connections: the server leaves the TCP connection open after sending a response. Subsequent requests and responses between the same client and server can be sent over the same connection.

Advantages of persistent connections:
1. A brand-new connection does not need to be established and maintained for each requested object — this avoids TCP buffer allocation and TCP variable overhead on both client and server, reducing the burden on the Web server.
2. With pipelining, requests for objects can be made back-to-back without waiting for replies to pending requests, which significantly reduces latency.
3. An entire Web page (e.g., the base HTML file and 10 images) can be sent over a single persistent TCP connection; even multiple Web pages from the same server can be sent over a single connection.

Disadvantages of non-persistent connections:
1. Each object suffers a delivery delay of two RTTs (one RTT for connection setup, one for request/response).
2. They place a significant burden on the Web server, which must allocate buffers and variables for each new connection.

The default mode of HTTP uses persistent connections with pipelining.

المرجع: Lecture 2 — Section 2.2.2

HTTP | Non-persistent | Persistent | Pipelining | ترجيح: high

Essay 11.

Explain the structure of an HTTP request message in detail, listing its main components.

ترجمة: اشرح بنية رسالة HTTP request بالتفصيل، مع ذكر المكونات الرئيسية.

An HTTP request message is written in ordinary ASCII text so that it can be read by humans. It consists of:
1. A request line (the first line), which has three fields: the method field, the URL field, and the HTTP version field. The method field can take on values including GET, POST, HEAD, PUT, and DELETE.
2. Header lines (subsequent lines), each followed by a carriage return and a line feed. Common header lines include Host, Connection, User-agent, and Accept-language.
3. An entity body (after the header lines and an additional carriage return/line feed). The entity body is empty with the GET method, but is used with the POST method to carry the user's input from a form.

Example request message:
GET /somedir/page.html HTTP/1.1
Host: www.someschool.edu
Connection: close
User-agent: Mozilla/5.0
Accept-language: fr

The Host: header specifies the host on which the object resides. Connection: close tells the server the browser wants the connection closed after sending the object. User-agent: identifies the browser type. Accept-language: indicates the user's preferred language.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Request | Request line | Headers | ترجيح: high

Essay 12.

Explain the structure of an HTTP response message in detail.

ترجمة: اشرح بنية رسالة HTTP response بالتفصيل.

An HTTP response message has three sections:
1. An initial status line (the first line), with three fields: the protocol version field, a status code, and a corresponding status message.
2. Six (or more) header lines, such as Connection, Date, Server, Last-Modified, Content-Length, and Content-Type.
3. An entity body, which is the meat of the message — it contains the requested object itself.

Example response message:
HTTP/1.1 200 OK
Connection: close
Date: Tue, 18 Aug 2015 15:44:04 GMT
Server: Apache/2.2.3 (CentOS)
Last-Modified: Tue, 18 Aug 2015 15:11:03 GMT
Content-Length: 6821
Content-Type: text/html
(data data data data data ...)

Connection: close means the server will close the TCP connection after sending the message. Date: is when the response was created/sent (not when the object was created). Server: identifies the server software. Last-Modified: indicates when the object was last changed and is critical for caching. Content-Length: is the byte count of the object. Content-Type: indicates the object's media type.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Response | Status line | Headers | ترجيح: high

Essay 13.

List and explain the common HTTP status codes given in this lecture.

ترجمة: اكتب واشرح أكواد الـ status الشائعة اللي ذكرت في المحاضرة.

Common HTTP status codes and their associated phrases:
1. 200 OK: The request succeeded and the information (the requested object) is returned in the response.
2. 301 Moved Permanently: The requested object has been permanently moved; the new URL is specified in the Location: header of the response message. The client software will automatically retrieve the new URL.
3. 400 Bad Request: A generic error code indicating that the request could not be understood by the server.
4. 404 Not Found: The requested document does not exist on this server.
5. 505 HTTP Version Not Supported: The requested HTTP protocol version is not supported by the server.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Response Message)

HTTP Status Codes ترجيح: high

Essay 14.

Compare the GET and POST HTTP methods, and briefly mention the HEAD, PUT, and DELETE methods.

ترجمة: قارن بين طريقتي GET و POST في HTTP، واذكر باختصار HEAD و PUT و DELETE.

GET method: used when the browser requests an object, with the requested object identified in the URL field. The entity body is empty with GET. It is the most common HTTP method.
POST method: an HTTP client often uses the POST method when the user fills out a form — for example, when a user provides search words to a search engine. With POST, the user is still requesting a Web page, but the specific content depends on what the user entered into the form fields. The entity body contains what the user entered in the form fields.
Important note: HTML forms do not necessarily use POST — they can use GET and include the inputted data in the requested URL (e.g., www.somesite.com/animalsearch?monkeys&bananas).

Other methods:
1. HEAD: similar to GET; often used by application developers for debugging.
2. PUT: often used in conjunction with Web publishing tools; allows a user to upload an object to a specific path (directory) on a specific Web server.
3. DELETE: allows a user or an application to delete an object on a Web server.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Methods GET POST HEAD PUT DELETE **ترجيح:** high

Essay 15.

List the five important network applications covered in this chapter and briefly state why each is included.

ترجمة: اكتب التطبيقات الخمسة المهمة اللي الكتاب بيركز عليها في الفصل وإيه سبب إدراج كل واحد.

The book covers five important network applications:
1. The Web — enormously popular; its application-layer protocol, HTTP, is straightforward and easy to understand.
2. Electronic mail — the Internet's first killer application; more complex than the Web because it uses several application-layer protocols.
3. DNS (directory service) — provides a directory service for the Internet; users invoke DNS indirectly through other applications; illustrates how core network functionality (name-to-address translation) can be implemented at the application layer.
4. P2P file sharing applications.
5. Video streaming on demand, including distributing stored video over content distribution networks.

المرجع: Lecture 2 — Section 2.1.6

Network Applications ترجيح: medium

Essay 16.

Explain the role of the following HTTP request header lines: Host, Connection, User-agent, and Accept-language.

ترجمة: اشرح دور الـ header lines دي في HTTP request: Host, Connection, User-agent, Accept-language.

Host: www.someschool.edu — specifies the host on which the object resides. This may seem unnecessary because a TCP connection is already in place to the host, but the information provided by the host header line is required by Web proxy caches.
Connection: close — tells the server that the browser does not want to bother with persistent connections; it wants the server to close the connection after sending the requested object.
User-agent: Mozilla/5.0 — specifies the user agent, i.e., the browser type making the request. Here the user agent is a Firefox browser. This header is useful because the server can actually send different versions of the same object to different types of user agents.
Accept-language: fr — indicates that the user prefers to receive a French version of the object, if such an object exists on the server; otherwise, the server should send its default version. Accept-language is one of many content negotiation headers available in HTTP.

المرجع: Lecture 2 — Section 2.2.3 (HTTP Request Message)

HTTP Headers Host Connection User-agent Accept-language ترجيح: medium

Calculation 1.

(Calculation) Calculate the total response time for fetching a base HTML file using non-persistent HTTP. Given: RTT = 100 ms, transmission time of the HTML file at the server = 50 ms. Ignore all other delays.

ترجمة: (مسألة) احسب إجمالي زمن الاستجابة لجلب base HTML file باستخدام non-persistent HTTP.
المعطيات: RTT = 100 ms، زمن نقل الـ HTML على السيرفر = 50 ms. تجاهل باقي التأخيرات.

Given:
- RTT (round-trip time) = 100 ms
- Transmission time of the HTML file at the server = 50 ms

Formula (from lecture): Total response time = 2 RTTs + transmission time at the server.

Step-by-step reasoning:
1. The first two parts of the three-way handshake take 1 RTT (100 ms).
2. After completing the first two parts of the handshake, the client sends the HTTP request message combined with the third part of the three-way handshake (the acknowledgment). Once the request arrives at the server, the server sends the HTML file. This HTTP request/response eats up another RTT (100 ms).
3. The server transmits the HTML file, which takes the transmission time (50 ms).

Total response time = 2 * 100 ms + 50 ms = 200 ms + 50 ms = 250 ms.

Answer: 250 ms.

المرجع: Lecture 2 — Section 2.2.2 (response time formula)

Calculation | RTT | Non-persistent | ترجيح: high

Calculation 2.

(Calculation) How many TCP connections are needed to fetch a Web page consisting of a base HTML file and 10 JPEG images over non-persistent HTTP? All objects reside on the same server.

ترجمة: (مسألة) كام TCP connection محتاج عشان تجيب صفحة ويب فيها base HTML + 10 صور JPEG باستخدام non-persistent HTTP؟ كل الـ objects على نفس السيرفر.

Given:
- Web page consists of 1 base HTML file + 10 JPEG images = 11 objects total.
- All objects reside on the same server.
- Connection mode: non-persistent HTTP.

In non-persistent HTTP, each TCP connection transports exactly one request message and one response message, and the connection is closed after the server sends the object.

Therefore, the number of TCP connections = number of objects = 11.

Answer: 11 TCP connections are generated when the user requests this Web page.

Note: This is exactly why the lecture introduces persistent connections — to avoid opening a new TCP connection for each object.

المرجع: Lecture 2 — Section 2.2.2 (Non-persistent walk-through)

Calculation | TCP connections | Non-persistent | ترجيح: high

.Calculation 3

(Calculation) Calculate the total time to fetch a Web page with 1 base HTML file and 9 JPEG images (10 objects total) using non-persistent HTTP, assuming serial TCP connections (one connection at a time). Given: RTT = 200 ms, transmission time per object = 20 ms. Ignore other delays.

ترجمة: (مسألة) احسب إجمالي الزمن لجلب صفحة ويب فيها 1 HTML + 9 JPEG (10 objects) باستخدام non-persistent HTTP مع اتصالات serial. المعطيات: RTT = 200 ms، زمن النقل لكل object = 20 ms.

Given:
- Number of objects = 1 HTML + 9 JPEG = 10 objects.
- RTT = 200 ms.
- Transmission time per object = 20 ms.
- Non-persistent HTTP, serial (one TCP connection at a time).

Formula per object: Time per object = 2 RTTs + transmission time.

Time per object = 2 * 200 + 20 = 400 + 20 = 420 ms.

Total time = number of objects * time per object = 10 * 420 = 4200 ms = 4.2 seconds.

Answer: 4200 ms (4.2 seconds).

Note: If the browser opens parallel TCP connections instead of serial, the actual wall-clock time could be shorter — but the total connection work and total bytes transferred remain the same.

المرجع: Lecture 2 — Section 2.2.2 (response time formula + non-persistent walk-through)

Calculation RTT Non-persistent ترجيح: high

Calculation 4.

(Calculation) Using persistent HTTP with pipelining, calculate the total time to fetch the same Web page (1 base HTML file + 9 JPEG images, 10 objects total). Given: RTT = 200 ms, transmission time per object = 20 ms. Assume requests are pipelined after the base HTML is received.

ترجمة: (مسألة) باستخدام persistent HTTP مع pipelining، احسب إجمالي الزمن لجلب نفس الصفحة (1 HTML + 9 JPEG = 10 objects). المعطيات: RTT = 200 ms، زمن النقل لكل object = 20 ms. افترض إن الطلبات بتتحط pipelined بعد ما HTML بيوصلك.

Given:
- 10 objects (1 base HTML + 9 JPEG).
- RTT = 200 ms.
- Transmission time per object = 20 ms.
- Persistent HTTP with pipelining.

Steps:
1. TCP three-way handshake: the first two parts take 1 RTT = 200 ms.
2. After the handshake, the client sends the HTTP request for the base HTML file over the same connection. The server returns the HTML file. This request/response takes 1 RTT + transmission time of the HTML = 200 + 20 = 220 ms.
3. With pipelining, after the HTML is received, the client sends requests for the 9 JPEG objects back-to-back over the same TCP connection (no new handshakes needed). The server sends them back-to-back.
   - One additional RTT for the 9-image request/response batch = 200 ms.
   - Transmission time for 9 images = 9 * 20 = 180 ms.

Total time = 200 (handshake) + 220 (HTML round) + 200 (images request/response) + 180 (images transmission) = 800 ms.

Simplified rough estimate (treating all 10 objects as one pipelined batch after the handshake):
Total ≈ 2 RTTs + sum of all transmission times = 2 * 200 + 10 * 20 = 400 + 200 = 600 ms.

Answer: between 600 ms and 800 ms, roughly 600 ms (compared to 4200 ms for non-persistent serial — a ~7x speedup).

المرجع: Lecture 2 — Section 2.2.2 (Persistent connections)

Calculation RTT Persistent Pipelining **ترجيح: high**

Calculation 5.

(Calculation) For a Web page with 1 base HTML file + 4 JPEG images (5 objects total), calculate the total time over non-persistent HTTP (serial, one TCP connection at a time). Given: RTT = 150 ms, transmission time per object = 30 ms.

ترجمة: (مسألة) لصفحة ويب فيها 1 HTML + 4 JPEG (5 objects)، احسب إجمالي الزمن على non-persistent HTTP (serial). المعطيات: RTT = 150 ms، زمن النقل لكل object = 30 ms.

Given:
- 5 objects (1 HTML + 4 JPEG).
- RTT = 150 ms.
- Transmission time per object = 30 ms.
- Non-persistent HTTP, serial.

Formula per object: Time = 2 RTTs + transmission time = 2 * 150 + 30 = 300 + 30 = 330 ms.

Total time (serial, 5 objects) = 5 * 330 = 1650 ms = 1.65 seconds.

Answer: 1650 ms (1.65 seconds).

Comparison: over persistent HTTP with pipelining, the same page would take roughly 2 RTTs + 5 transmission times = 2 * 150 + 5 * 30 = 300 + 150 = 450 ms — about 3.7x faster than non-persistent serial.

المرجع: Lecture 2 — Section 2.2.2 (Non-persistent walk-through)

Calculation | RTT | Non-persistent | ترجيح: high

Calculation 6.

(Calculation) An Internet telephony application sends 200 packets per second. Each packet is 200 bytes. Calculate the data rate at which the sender pumps data, and explain why this is acceptable for UDP but problematic for TCP.

ترجمة: (مسألة) تطبيق إنترنت تليفوني بيبعت 200 باكت في الثانية، كل باكت 200 بايت. احسب معدل ضخ البيانات، واشرح ليه ده مقبول لـ UDP ومشكلة لـ TCP.

Given:
- 200 packets/second.
- Each packet = 200 bytes.

Calculation:
Total data rate = 200 packets/s * 200 bytes/packet = 40,000 bytes/s = 40 KB/s = 320 kbps (since 1 byte = 8 bits).

Explanation:
UDP has no congestion-control mechanism, so the sender can pump data at this rate (320 kbps) regardless of the network state. Internet telephony is designed to tolerate some loss, so it works fine.
If TCP were used instead, TCP's congestion-control mechanism would throttle the sender when the network is congested between sender and receiver. This throttling causes additional latency and retransmission that is undesirable for time-sensitive applications like Internet telephony.
Therefore, developers of Internet telephony applications prefer UDP for lower latency, circumventing TCP's congestion control. They use TCP as a backup only when firewalls block UDP traffic.

المرجع: Lecture 2 — Section 2.1.4 (UDP Services + Figure 2.5 commentary)

Calculation UDP TCP Congestion Control ترجيح: medium